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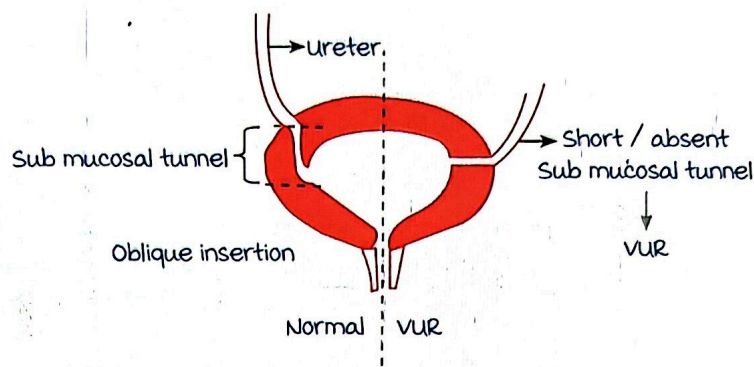
VUR AND UTI MANAGEMENT GUIDELINES

Vesico - Ureteric Reflux (VUR)

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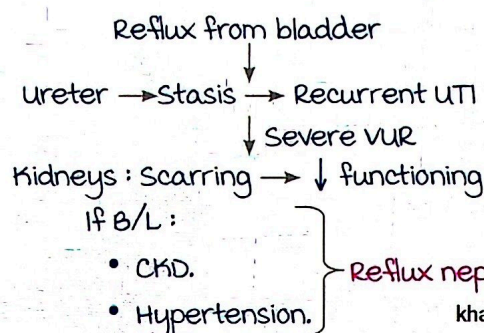
- Reflux of urine from bladder into ureter (usually congenital).

PATHOGENESIS :



Inheritance : **Autosomal dominant** → Screen siblings.

FEATURES:

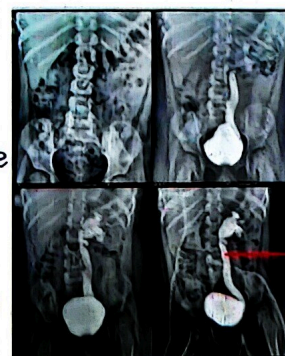


- VUR : m/c anomaly in a child with recurrent UTI.
- VUR + Bowel bladder dysfunction (BBD).
 - ↑ severity of VUR.
 - Delay in resolution of VUR.

DIAGNOSIS :

micturating cysto urethrogram (MCV) :

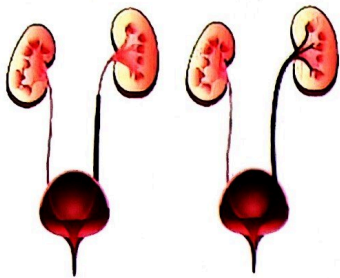
- MCV → Confirms diagnosis
- Can evaluate severity of disease
- AKA voiding cysto urethrogram (VCUG).



mcu

Retrograde flow
of contrast into
kidneys

Grading of VUR :



Grade 1 : Partial reflux
Grade 2 : Complete reflux



Grade 2 (+) :
Associated dilatation of
bladder & renal pelvis



Grade 3 (+) :
Tortuosity of ureter



Grade 4 (+) :
Distorted pelvi-calyceal
impressions

Radionuclide cystogram :

- For follow up of VUR.

Advantage : Less radiation exposure.

Disadvantage : Accurate grading not possible.

MANAGEMENT :

VUR resolves spontaneously.

1. Supportive management:

- Adequate fluids.
- mx of BBD.
 - Regular and complete voiding.
 - Relieve constipation if present.

2. Prevention of UTI $\xrightarrow{\text{If UTI occurs post prophylaxis}}$ Break through UTI : 3. Surgical correction

- Prophylactic antibiotics until BBD resolves.
- Indication : Grade 3-5 VUR.
- Drugs : Cotrimoxazole/nitrofurantoin/Cephalexin.
- Regimen : Until 2 years of age.

+

1 year after last episode of UTI.

ureteric re-implementation

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UTI

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SIGNIFICANCE OF UTI IN CHILDREN :

Presentation : Fever without a focus (Especially seen in infants).

Classification : Simple and complicated

UTI < 5 years : Investigate for $\rightarrow$ Renal scarring
 $\rightarrow$ Association with anomalies in urinary tract

m/c organism in UTI: *E. coli*.

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DEFINITIONS :**Complicated UTI :**

- High-grade fever ($>39^{\circ}\text{C}$).
- Systemic toxicity.
- Renal angle tenderness.
- $\uparrow$ S. creatinine.

Simple UTI :

- Low-grade fever.
- Dysuria.
- $\uparrow$ urine frequency.

Recurrent infection :

- ≥ 2 episodes of UTI
(Look for anomalies).

Significant Bacteriuria :

- **urine culture** : Definitive diagnosis of UTI.

Assessment of significant bacteriuria : By assessing CFU.

method of collection	Colony count (CFU/mL)	Probability of infection
Suprapubic aspiration	$> 10^3$	99%
urethral catheterization	$> 10^4$	95%
midstream clean catch (m/c)	$> 10^5$	90-95%

CFU : Colony Forming Units.

Asymptomatic bacteriuria: $\rightarrow$ Significant bacteriuria without any symptoms
 $\rightarrow$ **No treatment** required

Leukocyturia :

- $> 10 \text{ WBC/mm}^3$ in uncentrifuged urine sample OR $> 5 \text{ WBC/HPF}$ (High Power Field) in centrifuged urine sample.
- Not diagnostic of UTI.

MANAGEMENT :**Antibiotic management :**

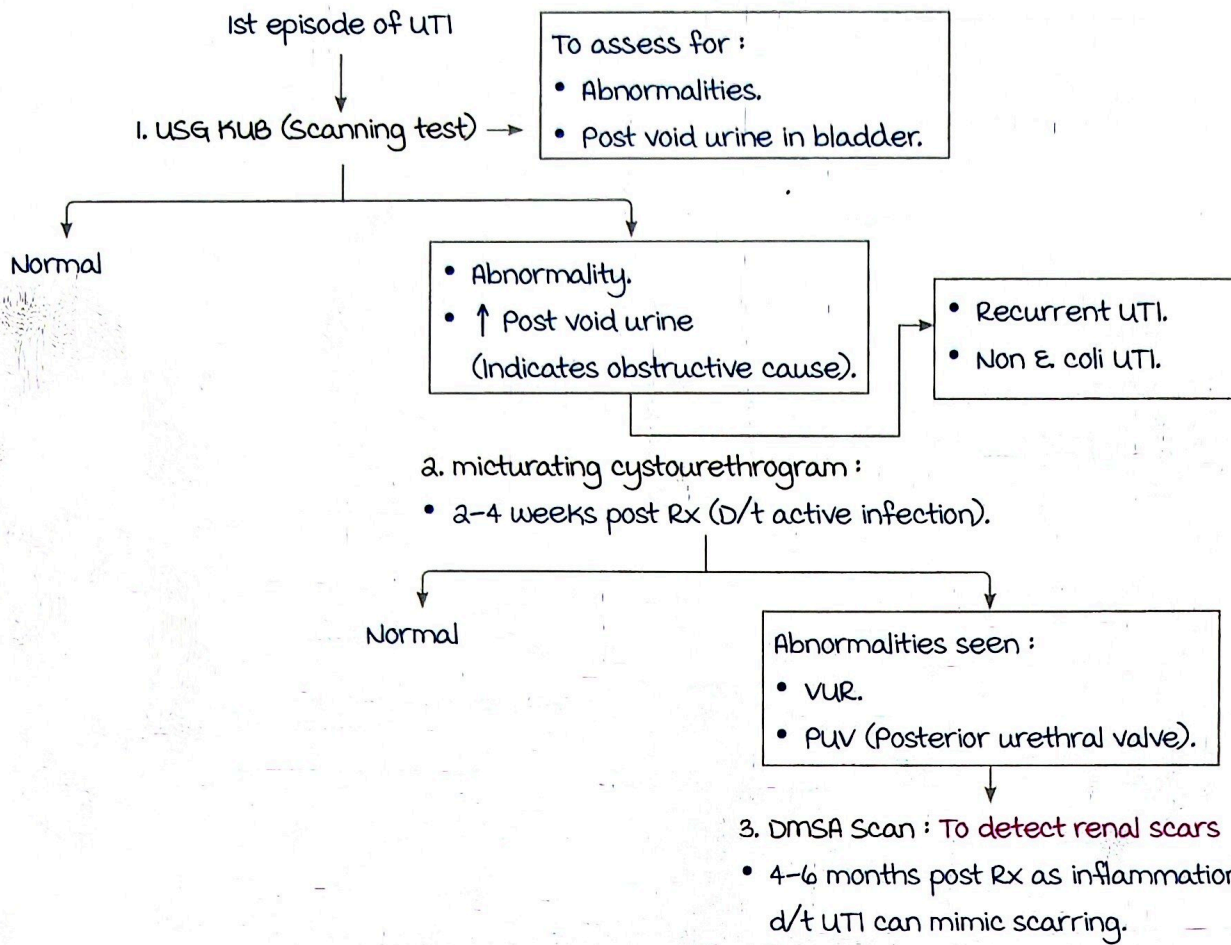
Requirement of hospitalization :

- Complicated UTI.
- Young infants (< 3 months).

Duration of treatment :

- Complicated UTI : 10-14 days $\rightarrow$ 3rd generation cephalosporin : Ceftriaxone/Cefotaxime
 Aminoglycoside (Single daily dose) : Amikacin/gentamicin
- Simple UTI : 7-10 days $\rightarrow$ Co-amoxiclav/Cefixime.

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